

Lift System

There are 8 modules in total.

Module 1 - Display the Position of Lift

Display the current floor positions of all lifts.

Sample Output:

```
Lift : L1 L2 L3 L4 L5
Floor : 0 0 0 0 0
```

Module 2 - Assign Lift to the Users

Assign a lift based on the user's current floor and destination floor.

Input:

```
2 5
(Source Floor: 2, Destination Floor: 5)
```

Output:

```
L1 is assigned
Lift : L1 L2 L3 L4 L5
Floor : 5 0 0 0 0
```

Module 3 - Assign Nearest Lift by Comparing Current Positions

Find and assign the nearest available lift to the user's source floor.

Assume:

```
Lift : L1 L2 L3 L4 L5
Floor : 5 2 7 9 0
```

Input:

```
4 10
(Source Floor: 4, Destination Floor: 10)
```

Output:

```
L1 is assigned
Lift : L1 L2 L3 L4 L5
Floor : 10 2 7 9 0
```

Explanation: L1 is nearest to floor 4 (L1 is at 5, difference = 1).

Module 4 - Assign Lift with Same Direction as User's Requirement

If two lifts are equally nearest to the user's source floor, assign the lift that is moving in the same direction as the user's request.

Example:

If the user wants to move from floor 4 to floor 2 (downward), and L3 is at floor 5 and L5 is at floor 3, then assign L3 because L3 will move downward from floor 5, which matches the user's downward direction.

Module 5 - Restrict Lifts to Specific Floor Ranges

Each lift is restricted to operate within a specific floor range:

- L1 and L2 : Floors 0 to 5 only
- L3 and L4 : Floors 6 to 10 only
- L5 : Floors 0 to 10 (all floors)
- Initially, all lifts are at floor 0.

Module 6 - Assign Lift with Least Number of Stops

When multiple lifts can serve the user, assign the one that will make the fewest stops to reach the destination.

Example:

L3 is at floor 9, L5 is at floor 8.

User wants to move from floor 8 to floor 0.

L3 will stop at: 8, 7, 6, then 0 -> NumberOfStops = 3

L5 will stop at: 8, 7, 6, 5, 4, 3, 2, 1, 0 -> NumberOfStops = 8

Assign L3 because it has fewer stops.

Module 7 - Assign Capacity to Lifts

Each lift has a maximum capacity (number of people it can carry at once). Assign lifts to users only if the lift has available capacity.

Module 8 - Handle Lift Under Maintenance

If a lift is under maintenance, its current floor position is marked as -1. Such a lift must never be assigned to any user under any circumstance.